

Tren dan Inovasi Masa Depan Sistem Tenaga

Menuju Era Smart Grid dan Modernisasi Energi

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Transformasi Digital Energi

Era sistem tenaga sedang mengalami transformasi fundamental dari sistem tenaga tradisional menuju era digital yang pintar, fleksibel, dan berkelanjutan.

Sistem Tenaga Tradisional:

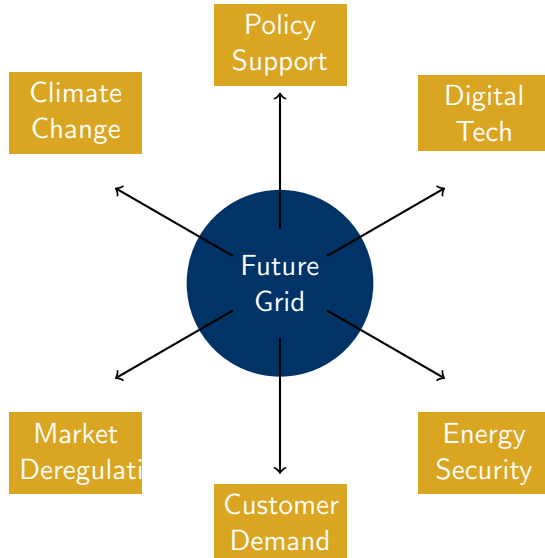
- Pembangkitan terpusat
- Arah aliran daya satu arah
- Operasi statis
- Monitoring manual

Sistem Tenaga Masa Depan:

- Pembangkitan terdistribusi
- Arah aliran daya dua arah
- Operasi adaptif
- Monitoring real-time

Fokus Utama: Implementasi di Provinsi Bengkulu

Driver Perubahan Sistem Tenaga



Definisi Smart Grid

Smart Grid adalah jaringan tenaga listrik yang menggunakan teknologi digital dan komunikasi canggih untuk meningkatkan efisiensi, keandalan, dan keamanan dalam produksi, transmisi, dan distribusi energi listrik.

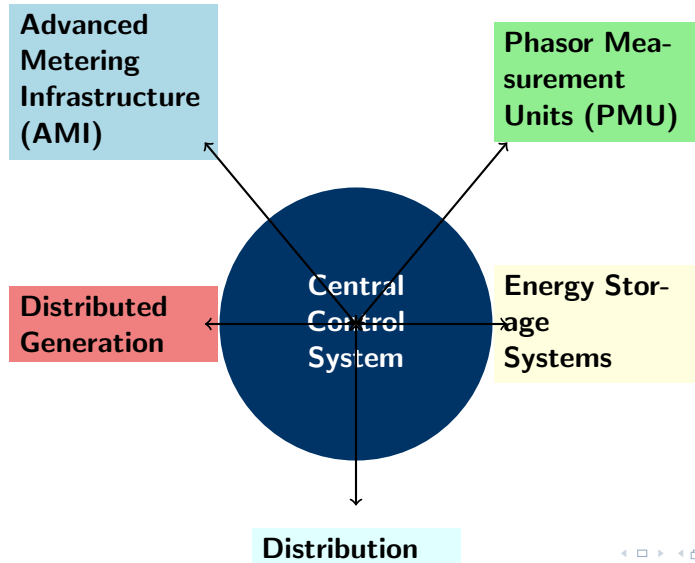
Karakteristik Smart Grid:

- **Two-way communication**
- **Automated control**
- **Self-healing**
- **Active consumers**
- **Asset optimization**

Benefits Smart Grid:

- Efisiensi energi meningkat 15-30%
- Rugi-rugi jaringan berkurang 20-30%
- Keandalan meningkat 25-40%
- Biaya operasi turun 15-25%

Komponen Utama Smart Grid



Advanced Metering Infrastructure (AMI)

Komponen AMI:

- **Smart Meters:** Pengukuran digital real-time
- **Communication Network:** PLC, RF, Cellular
- **Head End System (HES):** Data collection
- **MDMS:** Meter Data Management System
- **IT/OT Integration:** Business systems

Fungsi Utama AMI:

- Remote reading dan verification
- Two-way communication
- Load profiling dan analysis
- Time-of-use pricing
- Demand response
- Outage detection

Statistik Global AMI (2024)

- Global deployment: 72% pasar
- Asia-Pacific: 1,6 juta pengguna
- Indonesia: Pilot project
- Potential Bengkulu: 1,2 juta hogar

Phasor Measurement Units (PMU)

Prinsip Kerja PMU

PMU menggunakan sinyal GPS untuk sinkronisasi waktu dan mengukur phasor tegangan dan arus dengan presisi tinggi pada rate 30-120 sampel/detik.

Karakteristik Teknis PMU:

- Precision: ± 1 mrad angle accuracy
- Sampling rate: 30-120 Hz
- GPS timing: ± 1 s sync
- Frequency range: 0-360 Hz
- Dynamic range: 10:1 to 200:1

Aplikasi PMU dalam Smart Grid:

- Wide Area Monitoring (WAMS)
- Oscillation detection
- Accuracy: $\pm 1 \mu s$ timestamping
- Latency: < 20 ms data transmission
- Reliability: 99.999% uptime
- Dynamic line rating

Konsep Self-Healing

Self-healing grid adalah kemampuan sistem distribusi untuk secara otomatis mendeteksi, mengisolasi, dan memulihkan dari gangguan tanpa intervensi manual.

Fault Detection → Fault Isolation → Service Restoration → Normal Operation

Keunggulan Self-Healing Grid:

- Waktu pemulihan: dari jam menjadi menit (30-60)
- SAIDI reduction: 25-40%
- SAIFI reduction: 30-45%
- Biaya operasi: penurunan 15-25%

Tantangan Integrasi Energi Terbarukan

Tantangan Teknis Utama:

Intermitensi dan Variabilitas

- Fluktuasi output yang tidak terduga
- Perubahan kondisi cuaca
- Daily dan seasonal patterns

Grid Stability & Low Inertia

- Penurunan sistem inersia akibat Inverter-Based Resources (IBR)
- Perlu teknologi *Grid-Forming Inverters*
- Voltage Frequency stability

Parameter Teknis Kritis:

Ramping Rate & IBR Impact

- Solar PV: 70-90%/menit
- Penurunan inersia sintetis pada grid
- Grid requirement: $\geq 20\%$ /menit

Reserve Requirements

- Primary reserve: 10-15% capacity
- Secondary reserve: 5-10% capacity
- Fast Frequency Response (FFR) dari IBR

Tantangan Inersia Rendah pada Grid Masa Depan

Peningkatan penetrasi pembangkit berbasis inverter (PLTS PLTB) menurunkan inersia mekanis jaringan. Oleh karena itu, arsitektur kendali inverter berevolusi dari *Grid-Following* menjadi *Grid-Forming*.

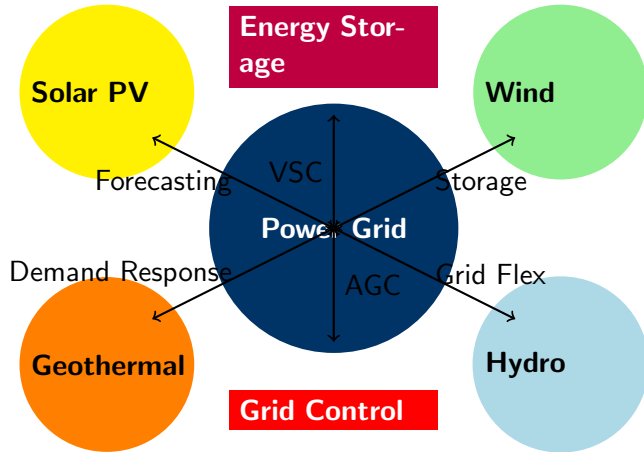
Grid-Following (GFL) Inverter:

- Membutuhkan sumber tegangan acuan (phase-locked loop / PLL).
- Bertindak sebagai sumber arus (*current source*).
- Rentang operasi terbatas pada jaringan lemah (*weak grid*).

Grid-Forming (GFM) Inverter:

- Membentuk amplitudo dan frekuensi tegangan sendiri.
- Bertindak sebagai sumber tegangan (*voltage source*).
- Menyediakan inersia sintetis dan *Fast Frequency Response* (FFR).

Solusi Integrasi Energi Terbarukan



Potensi Energi Terbarukan di Bengkulu

Potensi Surya (Solar):

- Kapasitas terpasang: 3.475 MW
- Irradiance: 5,9-8,3 jam/hari
- Faktor kapasitas: 23-35%
- Lokasi optimal: Coastal areas

Potensi panas Bumi (Geothermal):

- Potensi resource: 1.360 MW
- Lokasi: South-west region
- Teknologi: Flash + Binary

Potensi Air (Hydro):

- Total hydro: 945 MW
- Micro-hydro: 200 MW additional
- Stream characteristics: High head, low flow

Potensi Laut (Ocean Energy):

- Ocean waves: Undeveloped potential
- Ocean thermal: Experimental phase
- Research opportunities

Total Potensi: 6 GW renewable capacity

Perbandingan Teknologi ESS

Technology	Efficiency (%)	Power (MW)	Energy (MWh)	Cycle Life (cycles)	Cost (\$/kWh)
Lithium-ion	85-97	0.1-100	0.5-1,000	1,000-10,000	200-400
Flow Battery	75-85	1-100	4-1,000	10,000-20,000	500-1,000
Pumped Hydro	70-85	100-5,000	1-10,000	30,000+	50-150
CAES	70	100-1,000	2-10,000	30,000+	100-200
Flywheel	90-95	0.1-20	0.005-0.5	100,000+	5,000-15,000
Hydrogen	30-50	10-10,000	Unlimited	20,000+	200-500

Aplikasi ESS dalam Sistem Tenaga

Aplikasi Grid-Scale:

Frequency Regulation

- Response time: ≤ 100 ms
- Power range: 1-10 MW
- Economic benefit: \$2-5/MWh

Peak Shaving

- Load duration: 1-4 hours
- Power capacity: 10-100 MW
- Cost saving: \$15-30/MWh

Grid Stabilization

- Voltage support

Aplikasi Distributed:

Residential

- Solar integration: 2-10 kWh
- Backup power: 1-5 kWh
- Cost: \$300-800/kWh

Commercial

- Demand charge reduction
- Grid services: 50-500 kWh
- Payback: 5-10 tahun

Vehicle-to-Grid (V2G)

- EV fleet potential: 10-100 kWh

ESS Sizing untuk Aplikasi PLTS

Formula Perhitungan ESS

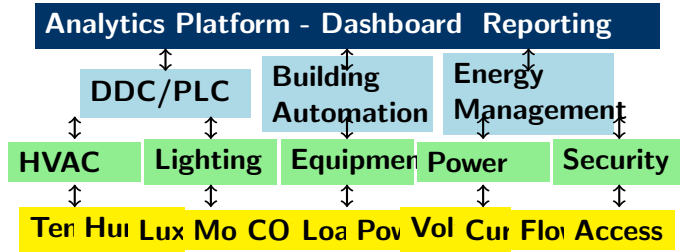
Untuk aplikasi peak shaving PLTS, sizing ESS dapat dihitung:

$$E_{ESS} = \frac{P_{peak} \times t_{duration}}{\eta_{ESS} \times DOD} \quad (1)$$

$$P_{ESS} = \frac{P_{solar}}{\eta_{ESS}} \quad (2)$$

dimana:

- E_{ESS} = Kapasitas energi ESS (kWh)
- P_{ESS} = Kapasitas daya ESS (kW)
- P_{peak} = Daya peak PLTS (kW)
- $t_{duration}$ = Durasi discharge (jam)
- η_{ESS} = Efisiensi round-trip (85-95%)



Optimasi dan Kontrol BEMS

Strategi Kontrol BEMS:

Model Predictive Control (MPC)

- Prediksi beban 24 jam ke depan
- Optimasi setpoint HVAC
- Integration dengan weather forecast
- Energy savings: 15-25%

Demand Response (DR)

- Automated load shedding
- Peak demand reduction
- Grid services participation
- Cost reduction: 10-30%

Machine Learning Control

Example

Contoh Implementasi UNIB Gedung Pascasarjana UNIB

- Energy consumption: 150 MWh/bulan
- BEMS cost: \$50.000
- Savings potential: 20-30%
- Payback period: 3-5 tahun
- ROI: 45-60%

Potensi Penghematan Tahunan

- Energy cost: \$8.000

Dampak Kendaraan Listrik terhadap Grid

Proyeksi Adopsi EV Indonesia:

Perkembangan Pasar EV

- 2020: 125 unit (pertama di Indonesia)
- 2024: 43.188 unit terjual
- Target 2030: 2,5 juta unit/tahun
- Market share target: 20%

Impact pada Electricity Demand:

Load Projection

- 2025: 0,2% dari total demand
- 2030: 2,5% dari total demand
- 2050: 23% dari total demand
- Additional load: 45 TWh/tahun

Kebutuhan Infrastructure Charging:

Stasion Charging Requirements

- Residential: 2,3 juta unit charging
- Public: 65.000 unit (50kW)
- Fast charging: 15.000 unit (150kW+)
- Investment: \$3-5 miliar

Grid Impact per Vehicle

- Daily charging: 8-12 kWh
- Peak power: 7-22 kW
- Load diversity factor: 0,3-0,5
- Grid stress hours: 18:00-22:00

Vehicle-to-Grid (V2G) Technology

Konsep V2G

V2G adalah teknologi yang memungkinkan energi listrik mengalir dari baterai EV kembali ke grid ketika diperlukan, mengubah kendaraan sebagai sumber energi bergerak.

Manfaat V2G untuk Grid:

- **Frequency Regulation:** Response time ≤ 100 ms
- **Peak Shaving:** Load leveling capability
- **Backup Power:** Emergency supply
- **Grid Stability:** Voltage support
- **Deferred Investment:** Reduced infrastructure

Benefits untuk EV Owners:

- **Revenue stream:** \$200-500/tahun per EV

Example

Kasus V2G Fleet (100 EV)

Technical Specs:

- Average capacity: 50 kWh/vehicle
- Available for V2G: 40 kWh (80%)
- Response power: 7-11 kW/vehicle
- Total available: 700-1100 kW

HVDC untuk Interkoneksi Antar Pulau

Proyek HVDC Sumatera-Jawa

Koneksi Sumatera-Jawa adalah proyek strategis PLN untuk mentransmisikan daya dari Jambi-Sumatera Selatan ke Jawa dengan total:

- Panjang: 112 km (68 mile)
- Kapasitas: 3.000 MW
- Investment: Rp565,3 triliun (US\$36 miliar)
- Type: VSC-HVDC

Keunggulan HVDC:

- **No reactive power** transmission
- **Asynchronous** interconnection
- **Independent control** dari frekuensi
- **Lower losses** untuk jarak ≥ 500 km
- **Smaller right-of-way**

Application untuk Bengkulu:

- Bulk transmission dari PLTS coastal
- Interkoneksi dengan Jawa grid
- Renewable energy export
- Grid stability improvement

HVDC Technology Comparison

Parameter	LCC-HVDC	VSC-HVDC
Converter Technology	Thyristor-based	IGBT-based
Reactive Power	Consumes 50-60% of P	Self-controlled 0-100%
Commutation	Requires strong AC grid	No commutation needed
Black-start capability	Limited	Available
Power reversal	Reversible polarity	Fast polarity change
Cost	Lower converter cost	Higher converter cost
Suitable for	Strong grids, bulk power	Weak grids, renewables

Ancaman Siber terhadap Infrastruktur Energi

Types of Cyber Attacks:

Network-based Attacks

- DDoS (Distributed Denial of Service)
- Man-in-the-middle
- Network sniffing
- IP spoofing

System-based Attacks

- Malware injection
- SCADA compromise
- PLC hijacking
- Firmware tampering

Physical Attacks

Impact Assessment:

Potential Consequences

- Grid blackout
- Equipment damage
- Data theft
- Safety hazards
- Economic loss

Historical Incidents

- Ukraine 2015: 225.000 customers
- Ukraine 2016: Transmission grid

Mitigation Strategies

Defense in Depth Strategy

Layer 1: Physical Security

- Access control systems
- Surveillance monitoring
- Perimeter protection

Layer 2: Network Security

- Firewalls dan intrusion detection
- Network segmentation
- VPN untuk remote access

Layer 3: System Security

- Antivirus/anti-malware
- Patch management
- Configuration hardening

UU No. 30 Tahun 2009 tentang Ketenagalistrikan

Framework UU 30/2009:

Prinsip Dasar

- Ketenagalistrikan sebagai hajat hidup orang banyak
- Asas safety, reliability, dan efficiency
- Asas keterjangkauan harga
- Asas keterjangkauan layanan
- Asas berkelanjutan

Peran Pemerintah

- Kebijakan nasional ketenagalistrikan
- Pengaturan tarif dasar listrik
- Pemberian perizinan usaha ketenagalistrikan
- Pengawasan keselamatan instalasi

Penetrasi Energi Terbarukan:

Target 2025

- Mix energi terbarukan: 23%
- PLTS kapasitas: 6,5 GW
- PLTB kapasitas: 1,8 GW

Regulasi Pendukung

- Feed-in tariff mechanism
- Net metering policy
- Green certificate program
- Carbon pricing mechanism

Example

RUPTL 2021-2030 untuk Sumatera

Investment Plan: Rp 1.347 triliun untuk pembangunan infrastruktur ketenagalistrikan Sumatera

Komponen Investasi Utama:

- Generation: 35 GW (60% renewable)
- Transmission: 23.500 km jalur baru
- Distribution: 2,3 juta pelanggan baru
- HVDC: Sumatera-Jawa interconnection

Development untuk Bengkulu:

Coastal Solar Projects

- 1.200 MW PLTS capacity
- \$1,2 miliar investment
- 2025-2027 development

Prioritas Modernisasi Grid:

Smart Grid Implementation

- AMI deployment: 500.000 smart meters
- SCADA upgrade
- Distribution automation

Regulasi Distributed Generation

PLTS Atap Regulations:

Permen ESDM No. 26/2021

- Capacity limit: 500 kW (commercial)
- Maximum 15% of local capacity
- Net metering allowed
- Grid connection requirements

Net Metering Program

- Credit calculation: 1:1 ratio
- Excess credit compensation
- Bankability period: 3-8 tahun
- Installation standards

Environmental Regulations:

AMDAL Requirements

- Projects ≥ 50 MW: Full AMDAL
- Environmental impact assessment
- Community consultation
- Environmental permit

Emission Standards

- Thermal plants: $\text{SO}_x \leq 400 \text{ mg/Nm}^3$
- $\text{NO}_x \leq 750 \text{ mg/Nm}^3$
- Particulate $\leq 80 \text{ mg/Nm}^3$
- Mercury $\leq 0,03 \text{ mg/Nm}^3$

Phase 1 Goals

- 50.000 smart meters
- SCADA upgrade
- Basic automation

Phase 2 Goals

- 500 MW PLTS
- 50 MW ESS
- Full AMI coverage

Phase 3 Goals

- V2G integration
- BEMS widespread
- HVDC connection

Basic AMI

SCADA; [rectangle, fill=lightyellow, text=black] at (6,1) **Phase 2**
2028-2030

Advanced Grid

Storage; [rectangle, fill=lightcoral, text=black] at (10,1) **Phase 3**
2031-2035

Full Smart

Integration;

[rectangle, fill=lightblue, text=black] at (2,-1) \$150
million; [rectangle, fill=lightblue, text=black] at (6,-1) \$400
million; [rectangle, fill=lightblue, text=black] at (10,-1) \$600
million;

[circle, fill=red, text=white] at (2,0) ; [circle, fill=red, text=white] at (6,0) ; [circle, fill=red,
text=white] at (10,0) ;

Energy Storage Roadmap Bengkulu

Target ESS Deployment

2030 Target: 500 MW / 2.000 MWh total capacity untuk support renewable integration

Distribution per Technology:

- Li-ion BESS: 60% (300 MW/1.200 MWh)
- Flow batteries: 25% (125 MW/500 MWh)
- Pumped hydro: 15% (75 MW/300 MWh)

Location Strategy:

Economic Analysis:

Grid-Scale Deployment

- Near PLTS sites: 60%
- Substation integration: 30%
- Load centers: 10%

Investment Requirements

- Capital cost: \$400-500 million
- Annual O&M: \$15-25 million
- Revenue potential: \$50-80 million

Application Mix

Benefits Projection

Economic Impact Analysis

Investment Impact Bengkulu

Total Investment 2025-2035: \$1,15 miliar untuk modernisasi sistem tenaga

Economic Multiplier Effect:

- Direct employment: 1.500+ jobs
- Indirect employment: 2.500+ jobs
- Economic output: \$2,3 miliar
- Tax revenue: \$180 juta/tahun

Energy Cost Reduction:

Consumer Benefits

- Residential: 10-15% cost reduction
- Commercial: 15-25% cost reduction
- Industrial: 20-30% cost reduction

Environmental Benefits:

Carbon Emission Reduction

- Target: 50% reduction 2035
- Annual CO2 savings: 2,5 juta ton
- Air quality improvement
- Climate change mitigation

Challenge dan Mitigation Strategies

Technical Challenges:

Grid Integration Complexity

- Intermittency management
- Grid stability maintenance
- Power quality assurance
- Protection coordination

Infrastructure Requirements

- Advanced metering deployment
- Communication network upgrade
- SCADA modernization
- Cybersecurity enhancement

Economic & Social Challenges:

Financial Barriers

- High upfront investment
- Long payback periods
- Funding mechanisms
- Risk management

Human Resources

- Technical skill requirements
- Training programs
- Knowledge transfer
- Public awareness

Mitigation Strategies

Key Takeaways

1. Smart Grid Transformation

Implementasi Smart Grid di Bengkulu akan meningkatkan efisiensi 20-30%, mengurangi rugi-rugi 25-35%, dan meningkatkan keandalan sistem secara signifikan.

2. Energy Storage Critical Role

ESS dengan kapasitas 500 MW/2.000 MWh akan memungkinkan penetrasi energi terbarukan hingga 65% dengan grid stability yang terjamin.

3. BEMS untuk Building Efficiency

Implementasi BEMS di gedung-gedung Bengkulu dapat menghemat energi 15-30% dengan ROI 35-60% dalam 3-5 tahun.

4. Regulatory Framework Support

UU 30/2009, RUPTL PLN, dan regulasi ESDM memberikan framework yang kuat untuk transformasi energi di Bengkulu.

Strategic Recommendations

Immediate Actions (2025-2027):

- ① Pilot project AMI 5.000 rumah
- ② Upgrade SCADA dan automation
- ③ PLTS coastal feasibility study
- ④ BEMS pilot UNIB campus
- ⑤ Cybersecurity enhancement

Medium-term (2028-2030):

- ① Mass deployment AMI
- ② 500 MW PLTS dengan ESS
- ③ V2G pilot fleet
- ④ HVDC feasibility study

Long-term (2031-2035):

- ① Full Smart Grid deployment
- ② V2G widespread implementation
- ③ BEMS mandatory commercial
- ④ HVDC interconnection
- ⑤ Regional smart grid network

Critical Success Factors:

- Government commitment
- Industry partnership
- Technical expertise
- Financial resources
- Public acceptance

Visi 2035: Bengkulu sebagai Model Smart Grid Indonesia

- 65% renewable energy mix
- 40% reduction in carbon emissions
- 30% energy cost reduction for consumers
- 50
- 1.500+ new green jobs created
- Model for replication in Indonesia

“Dari Bengkulu untuk Indonesia: Membangun Masa Depan Energi yang Pintar, Berkelanjutan, dan Inklusif”

“Smart Grid bukan hanya tentang teknologi, tetapi tentang membangun masa depan yang lebih baik untuk generasi mendatang”

Terima Kasih Questions?

Contact Information:

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Resources:

Materials:

- Lecture notes tersedia online
- Lab exercises Octave
- Reading assignments
- Project guidelines

“Energi Masa Depan Dimulai dari Bengkulu”